FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 13-06-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

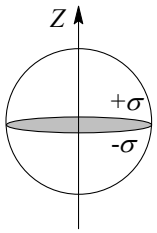
Physics

PART – I

SECTION – A
(One or More than one correct type)

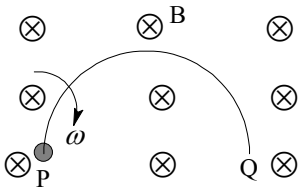
This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A hollow, insulating spherical shell has a surface charge distribution placed upon it, such that the upper hemisphere has a uniform surface charge density $+\sigma$, while the lower hemisphere has a uniform charge density $-\sigma$, as shown in the figure. Their interface lies in x-y plane. Which of the following statement(s) is/are correct?



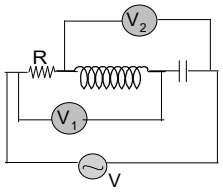
(A) The field at all points of x-y plane within the sphere points in the $-ve$ z-direction
(B) All points of the x-y plane within the sphere are equipotential
(C) The field at all points on z-axis outside the sphere point along positive z-direction
(D) The field at points on z axis which are on either side of origin outside the sphere is in opposite directions.

2. A wire is bent to form semi-circle of radius a. The wire rotates about its one end with angular velocity ω. Axis of rotation is perpendicular to plane of the semicircle. In the space, a uniform magnetic field of induction B exists along the axis of rotation as shown in figure. Then



(A) Potential difference between P and Q is equal to $2B\omega a^2$
(B) Potential difference between P and Q is equal to $2\pi^2 B\omega a^2$
(C) P is at higher potential Q
(D) P is at lower potential than Q

3. In an R-L-C series circuit shown in the figure, the readings of voltmeters V_1 and V_2 are 100V and 120V, respectively. The source voltage is 130V. For this situation, mark out the correct statement(s).



(A) Voltage across resistor, inductor and capacitor are $50V$, $50\sqrt{3}$ V and $120+50\sqrt{3}$ V, respectively
(B) Voltage across resistor, inductor and capacitor are $50V$, $50\sqrt{3}$ V and $120-50\sqrt{3}$ V, respectively
(C) Power factor of the circuit is $\frac{5}{13}$
(D) The circuit is capacitive in nature

4. An α-particle of mass m_α and q_α moves a velocity $V_0=\left(2\hat{i}+3\hat{j}-\hat{k}\right)ms^{-1}$ through a region, where both a uniform magnetic field and a uniform electric field. Which of the following statements is/are correct?

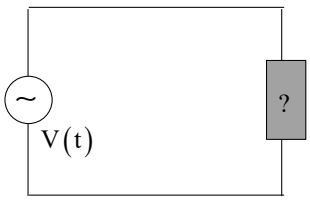
[Given, $B=\left(2i+4j+k\right)T$ and $E=\left(4i-j-2k\right)VM^{-1}$]

(A) Acceleration of α-particle is perpendicular to Z-axis
(B) Acceleration of α-particle is perpendicular to Z-axis
(C) de-Broglie's wavelength associated with α-particle is increasing with time
(D) de-Broglie's wavelength of α-particle keeps on decreasing with time

5. The polonium isotope $^{210}_{84}Po$ is unstable, it emits an α-particle with energy of 5.30MeV. The atomic mass of $^{210}_{84}Po$ is 209.9829 u and atomic mass of $^{4}_{2}He$ is 4.0026 u. Then,

(A) Daughter nucleus is $^{206}_{82}Pb$
(B) Disintegration energy of α-decay is 5.30MeV.
(C) Disintegration energy of α-decay is 5.40MeV.
(D) Mass of daughter nucleus is 205.9745u.

6. An AC voltage source is connected to a "black box" which contains a circuit, as shown in Figure. The elements in the circuit and their arrangement, however, are unknown. Measurements outside the black box provide the following information



V(t) = (80V) sin ωt
I(t) = (1.6A) sin (ωt+π/4)
Based on the above information we can infer:

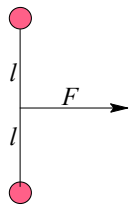
(A) The circuit is largely capacitative
(B) The circuit in the black box is in resonance
(C) The box must contain at least one resistor and one capacitor
(D) The average power delivered to the black box by the AC source is 45.25 W

SECTION – B
(Single Digit Integer Type)

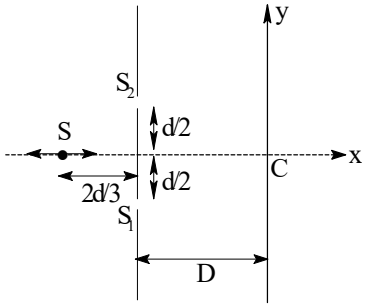
This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

7. Binding energy per nucleon of a fixed nucleus X^A is 6Me V. It absorbs a neutron moving with $KE=2MeV$, and converts into Y, emitting a photon of energy $1\ MeV$. If the Binding Energy per nucleon of Y (in MeV) is found to be $\frac{(xA-1)}{(A+1)}$. Find the value of x.

8. Two identical masses, placed on a smooth horizontal surface, are connected by a light string of length 2l. A constant force F starts acting on the system in the direction shown. After some time the masses collide inelastically. The energy lost in this collision is Fl/N. Find the value of N.

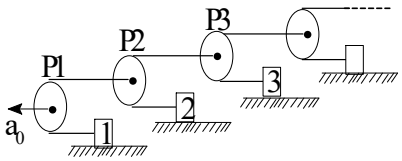


9. In Young's Double slit experiment, a symmetrically placed, long, linear and monochromatic source (S) of constant power is undergoing oscillations along x-axis with small amplitude A. Its mean position is shown in the figure. If I_0 is the intensity due to single slit near centre (C) of screen (y-z plane) when source is its mean position. If the amplitude of intensity variation ΔI_{max} at any of the bright fringe near centre is $\frac{8xAI_0}{25d}$, then find the value of x. [Assume D>>d]

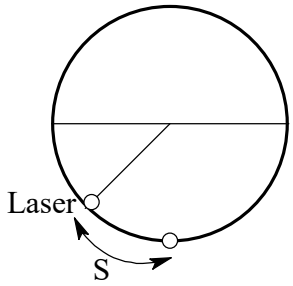


10. A spherical planet of radius r, painted black, travels around the sun in a circular orbit at a distance D from its centre $(r<<<D)$. The sun, a sphere of radius R, radiates as a blackbody at a temperature $T_0=6000K$ and subtends an angle 0.01 rad as seen from the planet. The equilibrium temperature T of the planet is $x\times100$. Find x

11. In the figure all pulleys (P1, P2, P3…….) are massless and all the blocks (1, 2, 3….) are identical, each having mass m. The system consists of infinite number of pulleys and blocks. Strings are light and in extension and horizontal surfaces are smooth. Pulley P1 is moved to left with a constant acceleration of a_0. The acceleration of block 1 will be (Assume the strings to remain horizontal) ka_0. Find 2k



12. A horizontal cylindrical tank $6\pi m$ in diameter is half full of water. The space above the water is filled with a pressurized gas of unknown refractive index. A small laser can move all the curved bottom of the water and aims a light beam towards the centre of the water surface. When the laser has moved a distance $S=1m$ or more (measured along the curved surface) from the lowest point in the water, no light enters the gas. If the refractive index of the gas is $(\mu_{max}=4/3)$ $2/\sqrt{n}$,then n is



SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

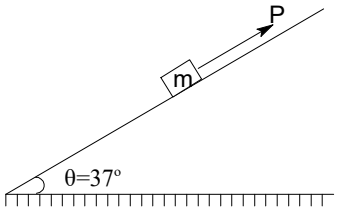
13. A particle moving with constant speed turns left by an angle of $74°$ after travelling ever 1m distance. If it returns back to its starting point after 18 second, then the magnitude of average acceleration of the particle from its initial position to just before it takes its second turn is $50\,k\,m/s^2$. Find the value of k ___

14. The electrostatic flux through a rectangular area of $\frac{7}{188}m^2$ lying in a plane $2x+3y+6z=10$ due to an electric field $\left(8\hat{i}+6\hat{j}+10\hat{k}\right)V/m$ will be in $(V\text{-}m)$.

15. A particle of mass equal to mass of the earth is projected by an external agency so that there is no recoil in earth when particle is projected. In this case escape velocity of the particle will be $\sqrt{(10k)gR_e}$, (g is acceleration due to gravity on the surface of the earth and R_e = radius of the earth). Then find the value of k ____

16. Steel has density $7800\,kg/m^3$ and will break if tensile stress exceeds $7.0\times10^8\ N/m^2$. A person wants to make a guitar string from 4.0 gm of this steel. The guitar string must be able to withstand a tension of 900 N. The maximum speed of transverse wave can be $x.xx\times100$ m/s

17. A block of mass m is being pulled up the rough incline, inclined at an angle $37°$ with horizontal by an agent delivering constant power P. The coefficient of friction between the block and the incline is μ. Find the maximum speed (in km/hr) of the block during the course of ascent.
[Take : P = 60 W, m = 1 kg, μ = 0.5]



18. The internal energy of a monatomic ideal gas is $1.5\,nRT$. One mole of helium is kept in a cylinder of cross-section 8.5 cm2. The cylinder is closed by a light frictionless piston. The gas is heated slowly in a process during which a total of 42 J heat is given to the gas. The temperature rises through $2°C$. The distance moved by the piston is given as a × 10^b m in scientific notation. Find the value of $a+b$. [Take R=25/3 in SI unit, atmospheric pressure = 100 kPa]

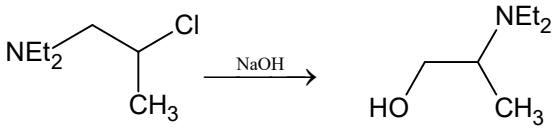
Chemistry

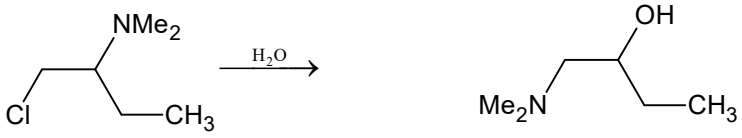
PART – II

SECTION – A
(One or More than one correct type)

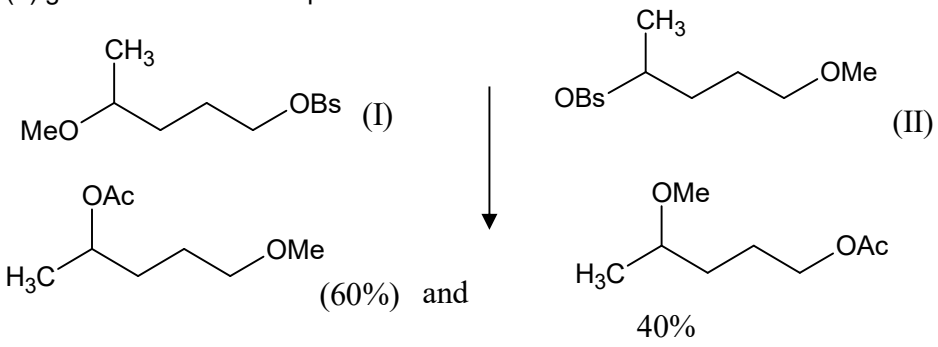
This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which is/are correct?

(A) 

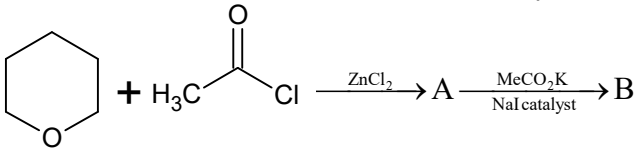
(B) 

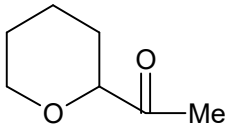
(C) Acetolysis of both 4-methoxy-1-pentyl-brosylate (I) and 5-methoxy-2-pentyl brosylate (II) gave same mixture of products.

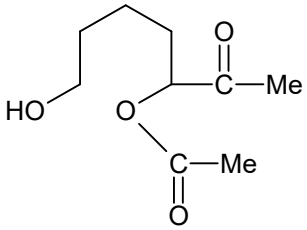


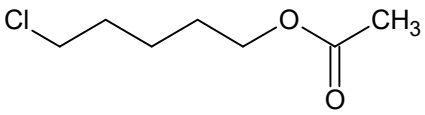
(D) Effectiveness of halogens as neighbouring group decreases in the order I > Br > Cl

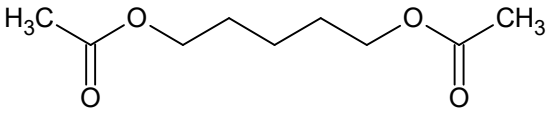
20. Which is/are correct about below reaction sequence?



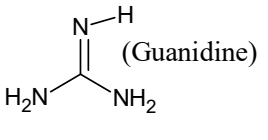
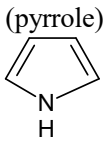
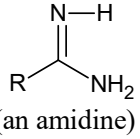
(A) Compound (A) is 

(B) Compound (B) is 

(C) Compound (A) is 

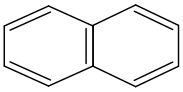
(D) Compound (B) is 

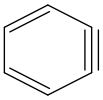
21. Which is/are wrong?

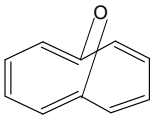
Compare the basic character of  (Guanidine)  (pyrrole) and  (an amidine)

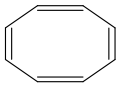
(A) Amidine >guanidine > pyrrole
(B) Pyrrole> guanidine> amidine
(C) Guanidine> amidine > pyrrole
(D) Amidine> pyrrole > guanidine.

22. Which of the following compounds exhibit aromatic character?

(A) 

(B) 

(C) 

(D) 

23. Which is/are correct?

(A) Except Sc, among 3d series transition metals all can show +2 oxidation state.
(B) Sc (I) is more stable than Sc(II) and Sc(III)
(C) Ti (IV) is more stable than Ti(III) and Ti(II)
(D) Ferrates $[FeO_4]^{2-}$ are not stable, so decompose to give FeO_3 and O_2.

24. Which catalyst and process is correctly matched.

(A) Decomposition of $KClO_3, MnO_2$
(B) Haber's process, Mo
(C) Lead chamber process, NO
(D) Contact process, V_2O_5

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. Find the total number specie(s) among the following containing $3c-2e$ bond.

$Be_2H_4, (BeH_2)_n, Be_2Cl_4, Al_2(CH_3)_6, Al_2Cl_6, I_2Cl_6, B_2H_6, B_2H_2(CH_3)_4$

26. Find the total number of compound(s), which is/are repelled by magnetic field.
$K_2O_2, KO_2, O_2[PtF_6], NO[BF_4], KCN, Na_2C_2$

27. Among the following, the number of ions/molecules having at least one atom sp hybridized is_____.
C_3S_2, NO_2^-, NO_2^+, benzene, benzyne, diazonium cation, $C_6H_5^+, C_3^{4-}, Cr_2O_7^{2-}, CaC_2, (SCN)_2$

28. Find the number of water of crystallization in molecule of butter of tin_____

29. What should be the ratio of the velocities of CH_4 and O_2 molecules so that they are associated with de-Broglie waves of equal wavelengths?

30. $0.5g$ of a mixture of K_2CO_3 and Li_2CO_3 requires 30 mL of 0.25 $N-HCl$ solution for neutralisation. The percentage composition of Li_2CO_3 in the mixture is.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The internal energy change in the conversion of 1.0 mole of the calcite form of $CaCO_3$ to the aragonite form is $+0.21kJ$. Calculate the enthalpy change (in J) when the pressure is 1.0 bar; given that the densities of the solids are $2.71g\,cm^{-3}$ and $2.93g\,cm^{-3}$ respectively.

32. Given $Ag^+(NH_3)_2 \rightleftharpoons Ag^+ + 2NH_3, K_c = 8.2\times10^{-8}$ and $K_{sp(AgCl)} = 2.378\times10^{-10}$ at 298K. Calculate the concentration of the complex in $1.0M$ aqueous ammonia_____.

33. K_p for the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at 400°C is 3.28. Calculate K_c____.

34. If 200 ml of $0.2M$ $BaCl_2$ solutions is mixed with 500 ml of 0.1 M Na_2SO_4 solutions. Calculate π of resulting solutions. (Assuming precipitate is completely precipitated)

35. If a completely charged lead storage cell is found to have 1.6 litre of 6M H_2SO_4 3.5 ampere of current is consumed for 6 hours, what is the concentration of sulphuric acid left if no change in volume is considered?

36. A mixture of ethane and ethene occupies 40 L at 1.0 atm, and 400K. The mixture reacts completely with 130g of O_2 to produce CO_2 and H_2O. Assuming ideal gas behaviour, calculate the mole fractions of C_2H_6 in the mixture.

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If in a right angled triangle, having integer sides the perimeter of triangle is equal to the area of triangle then possible area of triangle is/are
(A) 12
(B) 18
(C) 24
(D) 30

38. For the real values of λ, the curve $\lambda\left(x^2+y^2+2y+1\right)=\left(x-2y+3\right)^2$ represents
(A) an ellipse, if $\lambda > 5$
(B) an ellipse, if $0 < \lambda < 5$
(C) an hyperbola, if $\lambda > 5$
(D) an hyperbola, if $0 < \lambda < 5$

39. Let $\left(a_1,a_2,a_3.....a_n\right)$ be a permutation of $\left(1,2,3.....n\right)$ for which $a_1 > a_2 > a_3 > ..... > a_{\frac{n}{2}}$ and $a_{\frac{n}{2}} < a_{\frac{n}{2}+1} < a_{\frac{n}{2}+2} < ... < a_n$ for n as an even positive integer. Also $a_1 > a_2 > a_3..... > a_{\frac{n+1}{2}}$ and $a_{\frac{n+1}{2}} < a_{\frac{n+1}{2}+1} < ... < a_n$ for n as an odd positive integer. Let the total number of permutation of n be p(n), if $200 < p(n) < 500,$ then values of n is/are
(A) 10
(B) 11
(C) 12
(D) 13

40. Let a, b, c be three real numbers such that $1 \geq a \geq b \geq c \geq 0$. If λ (real or complex) is a root of the cubic equations $x^3 + ax^2 + bx + c = 0$ then
(A) $\lambda = 1$
(B) $|\lambda| > 1$
(C) $|\lambda| < 1$
(D) $\lambda = -1$

41. Let A, B, C be n×n real matrices and are pair wise commutative and $ABC = O_n$ and if $\lambda = \det(A^3 + B^3 + C^3)$. det (A + B + C) then
(A) $\lambda > 0$
(B) $\lambda < 0$
(C) $\lambda = 0$
(D) $\lambda \in (-\infty, \infty) - \{0\}$

42. Let f: A $\to$ A, where A is a set of non-negative integers and satisfying the condition.
(i) $x - f(x) = 19\,[x/19] - 90[f(x)/90], \forall\, x \in A$
(ii) $1900 < f(1900) < 2000$
then f(1900) equals, where [ ] indenter greatest integer function
(A) 1994
(B) 1900
(C) 1904
(D) 1890

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. If the end points A, B of the diameter of a circle are such that their abscissas are the roots of the equation $x^2 + ax - 16 = 0$ and their ordinates are the roots of $x^2 + 24x - 9 = 0$, If $a \in N$ and radius 'R' of the circle, with AB as diameter is an integer then, $R - a =$ ____

44. If the line $y = \sqrt{3}x$ cuts the curve $x^3 + y^3 = xy + 2$ at the points A, B, C. If O is the origin, then [OA. OB. OC] = _____ (where [.] denotes the greatest integer function).

45. In a Δ^{le} ABC, D is a point on BC such that $\frac{BD}{DC} = \frac{2}{3}$, point E lies on AB such that $\frac{BE}{EA} = \frac{2}{1}$, AD and CE intersect at F, $\frac{AF}{FD} = \frac{n}{n+1}$, $n \in N$. Then n = _____

46. The equations of four circles are $(x \pm (\sqrt{2} - 1))^2 + (y \pm (\sqrt{2} - 1))^2 = (\sqrt{2} - 1)^2$. Then the radius of a circle touching all the four circles internally is equal to _____

47. Let $a,b,c \in R^+$ and $432ab^2c^3 \geq (a+b+c)^6$. If $a + 2b + 3c = 6$, then $a + b^2 + c^3$ is equal to

48. The number of possible values of $\theta \in [0, 5\pi]$ so that $x^3 + (\tan\theta)\, x + 1 = 0$ has a repeated root is equal to

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $f(x) = \begin{cases} 2^{[x]} & ; x < 0 \\ 2^{[-x]} & ; x \geq 0 \end{cases}$, then the area bounded by y = f(x) and y = 0 is

50. If roots of equation $4\sqrt{6}x^3 - 51x^2 + 324\sqrt{6}x - 1296 = 0$ are the radii r_1, r_2, r_3 of the escribed circles of a triangle then semi–perimeter is

51. If n! has exactly (20) zeroes at the end, then number of such possible numbers is _____

52. If $y = f(x)$ is the solution of the differential equation $x(y^3 - x)dy = y(x + y^3)dx$ and $f(1) = (-2)^{1/3}$ and $f^{-1}(-2) = k$. Then $|k|$ = _____ (Where|.| denotes the modulus function)

53. Two circles of radius 1 and 4 touch each other externally. Another circle of radius r touches both circles externally and also one direct common tangent of the two circles. Then $\left[\frac{1}{r}\right]$ = _____ (where [.] donates greatest integer function)

54. ABC is an acute angle triangle, $\angle A = 30°$, H is the orthocentre and M is the midpoint of BC. On the line HM a point T is taken such that HM = MT. If BC = 4cm, then the length of AT is _____